AI-Assisted Coding
Lab Assignment - 10.4

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Lab 9 – Code Review and Quality: Using AI to Improve Code Quality and Readability

Task 1: AI-Assisted Syntax and Code Quality Review

Scenario

You join a development team and are asked to review a junior developer's Python script that fails to run correctly due to basic coding mistakes. Before deployment, the code must be corrected and standardised.

Expected Outcome

- Fully corrected and executable Python code
- AI-generated explanation describing:
    - Syntax fixes
    - Naming corrections
    - Structural improvements
- Clean, readable version of the script

Code With Errors:

```
#student management script

def calculate_average(marks)
def calculate_average(marks):
    total = 0
     for i in marks:
    for i in marks:
        total = total + i
    avg = total / len(mark)
    return Average
    avg = total / len(marks)
    return avg

def display_result(name, average):
print("Student Name:", Name)
    print("Student Name:", name)
    print("Average Marks:", average)

marks = [80, 75, 90, 85]

average = calculateAverage(marks)
average = calculate_average(marks)

display_result("Sushmitha", average)
```

Review:

✓ Optimizing tool selection...

✓ Reviewed task-1.py

I found several issues in the code. Let me fix them:

✓ Updated task-1.py

Fixed the following issues:

1. **Line 3**: Added missing colon after function definition
2. **Line 5**: Fixed indentation of `for` loop
3. **Line 7**: Changed `len(mark)` to `len(marks)`
4. **Line 8**: Changed `return Average` to `return avg`
5. **Line 11**: Changed `Name` to `name` and fixed indentation
6. **Line 15**: Changed `calculateAverage` to `calculate_average` to match the function name

The code should now run correctly and display the student's average marks.

Corrected Code:

```python
#student management script

def calculate_average(marks):
    total = 0
    for i in marks:
        total = total + i
    avg = total / len(marks)
    return avg

def display_result(name, average):
    print("Student Name:", name)
    print("Average Marks:", average)

marks = [80, 75, 90, 85]

average = calculate_average(marks)

display_result("Sushmitha", average)

```

Observation:
During the review, several syntax and naming errors were identified, including a missing colon, incorrect indentation, wrong variable references, and mismatched function names. These issues were corrected to follow proper Python syntax and naming conventions. After fixing them, the script executed successfully and produced the correct average marks output. The final version is clean, readable, and properly structured.

Task 2: Performance-Oriented Code Review
Scenario
A data processing function works correctly but is inefficient and slows down the system when large datasets are used
Expected Outcome
• Optimized duplicate-detection logic (e.g., using sets or hash-based structures)
• Improved time complexity
• AI explanation of performance improvement
• Clean, readable implementation.

Original Code:

```
def find_duplicates(data):
    duplicates = []

    for i in range(len(data)):
        for j in range(i + 1, len(data)):
            if data[i] == data[j] and data[i] not in duplicates:
                duplicates.append(data[i])

    return duplicates

numbers = [1, 2, 3, 4, 2, 5, 6, 3, 7, 8, 1]
print(find_duplicates(numbers))
```

Optimized Code:

```
#Optimized Code
def find_duplicates(data):
    seen = set()
    duplicates = set()

    for item in data:
        if item in seen:
            duplicates.add(item)
        else:
            seen.add(item)

    return list(duplicates)

numbers = [1, 2, 3, 4, 2, 5, 6, 3, 7, 8, 1]
print(find_duplicates(numbers))
```

Time Complexity Comparison

Version	Time Complexity
Original	$O(n^2)$
Optimized	O(n)

Why was the original approach inefficient?
• The original approach used nested loops, resulting in $O(n^2)$ time complexity.
• It checked every pair of elements in the list, leading to redundant comparisons.
How does the optimised version improve performance?
• The optimized version uses a set to track seen elements, achieving O(n) time complexity.
• It avoids redundant comparisons by only checking if an element has been seen before.

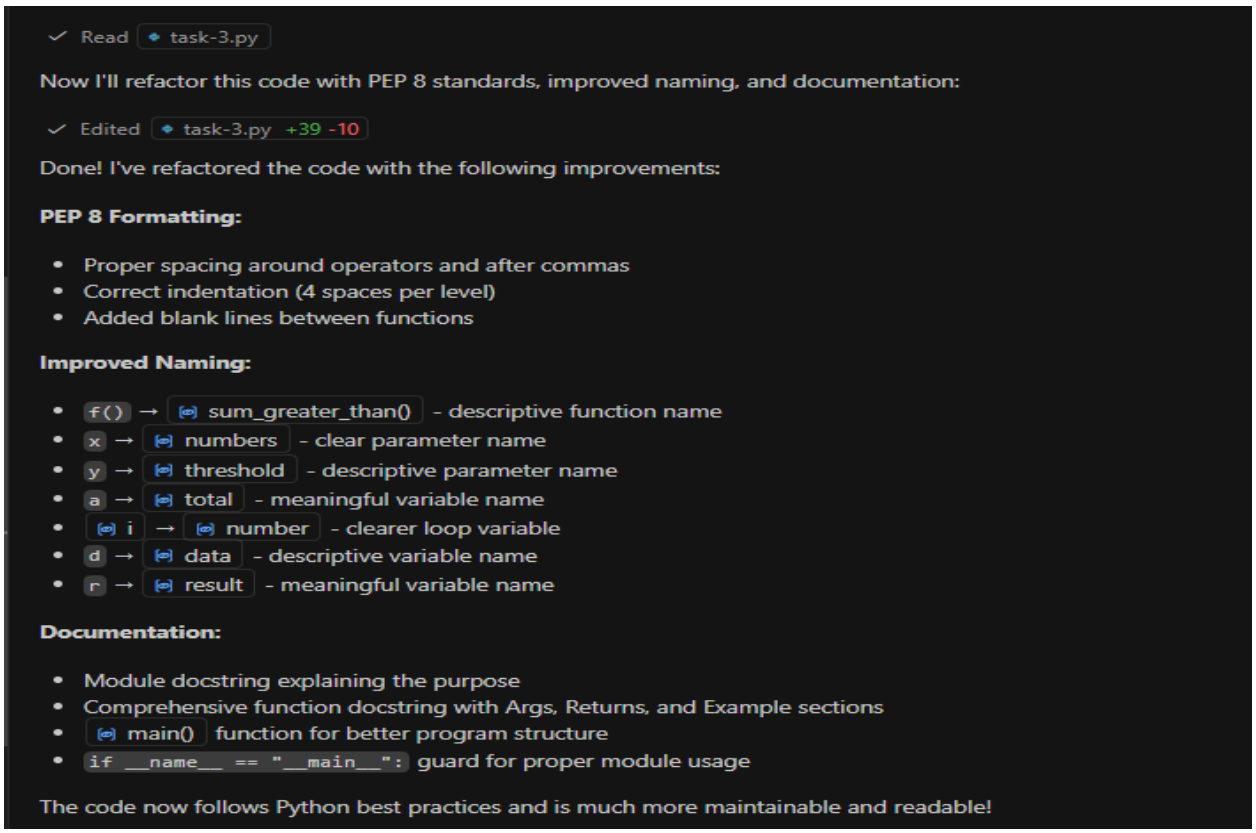
Observation:
The original duplicate detection function used nested loops, resulting in $O(n^2)$ time complexity, which makes it slow for large datasets. The performance bottleneck was caused by repeatedly comparing each element with every other element. By refactoring the code using a set for fast lookups, the time complexity was reduced to O(n). The optimised version improves efficiency while preserving the correctness of the output.

Task 3: Readability and Maintainability Refactoring
Scenario
A working script exists in a project, but it is difficult to understand due to poor naming, formatting, and structure. The team wants it rewritten for long-term maintainability.
Expected Outcome
• Clean, well-structured code
• Descriptive function and variable names
• Proper indentation and formatting
• Docstrings explaining the function's purpose
• AI explanation of readability improvements

Code:

```
def f(x,y):
 a=0
 for i in x:
  if i>y:
   a=a+i
 return a

d=[10,5,20,3,15]
r=f(d,10)
print(r)
```

Review:



✓ Read task-3.py

Now I'll refactor this code with PEP 8 standards, improved naming, and documentation:

✓ Edited task-3.py +39 -10

Done! I've refactored the code with the following improvements:

PEP 8 Formatting:

- Proper spacing around operators and after commas
- Correct indentation (4 spaces per level)
- Added blank lines between functions

Improved Naming:

- `f()` → sum_greater_than() - descriptive function name
- `x` → numbers - clear parameter name
- `y` → threshold - descriptive parameter name
- `a` → total - meaningful variable name
- i → number - clearer loop variable
- `d` → data - descriptive variable name
- `r` → result - meaningful variable name

Documentation:

- Module docstring explaining the purpose
- Comprehensive function docstring with Args, Returns, and Example sections
- main() function for better program structure
- `if __name__ == "__main__":` guard for proper module usage

The code now follows Python best practices and is much more maintainable and readable!

Refactored Code:

```python
"""
Module for calculating the sum of elements greater than a threshold.
This module provides functionality to filter and sum elements from a list
based on a minimum threshold value.
"""
def sum_greater_than(numbers, threshold):
    """
    Calculate the sum of all numbers greater than a given threshold.
    Args:
        numbers (list): A list of numeric values to filter and sum.
        threshold (int/float): The minimum value threshold for filtering.
    Returns:
        int/float: The sum of all numbers greater than the threshold.
    Example:
        >>> sum_greater_than([10, 5, 20, 3, 15], 10)
        35
    """
    total = 0
    for number in numbers:
        if number > threshold:
            total += number
    return total
def main():
    """Execute the main program logic."""
    data = [10, 5, 20, 3, 15]
    result = sum_greater_than(data, 10)
    print(result)

if __name__ == "__main__":
    main()
```

Observation:

The original script was difficult to understand due to unclear naming, poor formatting, and lack of documentation. After refactoring, meaningful function and variable names were introduced, and proper PEP 8 formatting was applied. Docstrings were added to clearly explain the purpose, parameters, and return values. The final version is clean, readable, and easier to maintain for long-term development.

Task 4: Secure Coding and Reliability Review

Scenario

A backend function retrieves user data from a database but has security vulnerabilities and poor error handling, making it unsafe for production deployment.

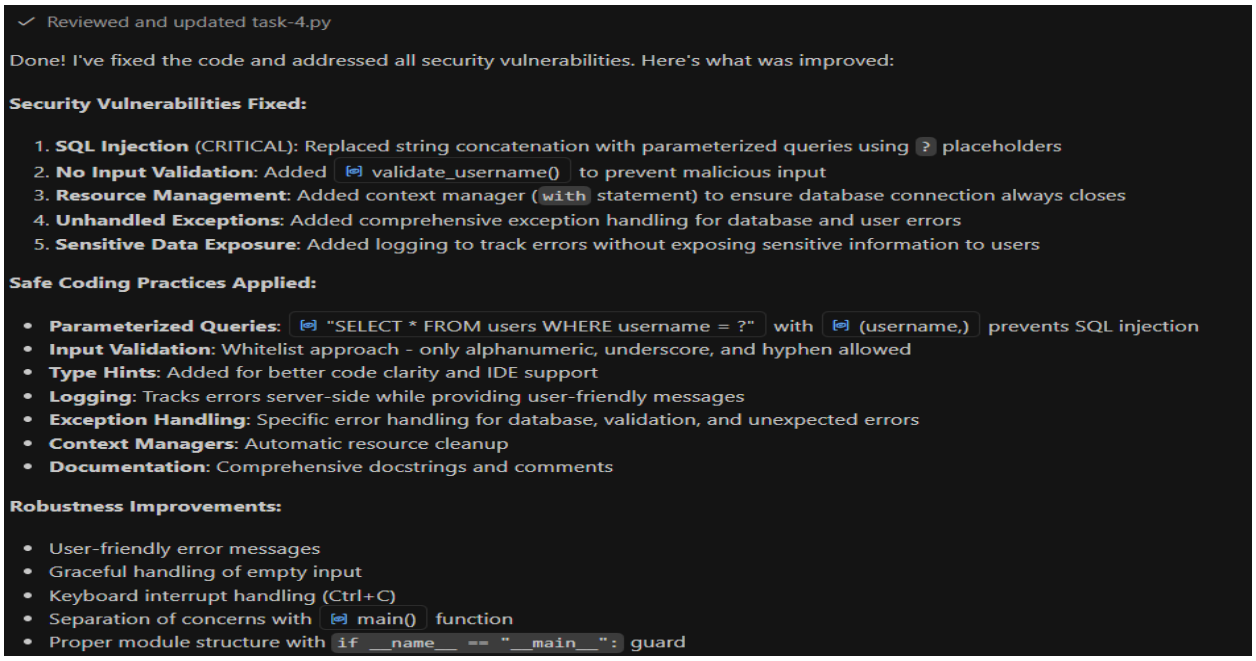
Expected Outcome

- Secure SQL queries using parameterized statements
- Input validation logic
- Try-except blocks for runtime safety
- AI-generated explanation of security improvements
- Production-ready code structure

Code:

```python
import sqlite3

def get_user(username):
    conn = sqlite3.connect("users.db")
    cursor = conn.cursor()

    query = "SELECT * FROM users WHERE username = '" + username + "'"
    cursor.execute(query)

    result = cursor.fetchone()
    conn.close()

    return result

user_input = input("Enter username: ")
print(get_user(user_input))
```

Review:



Refactored Code:

```python
import sqlite3
import logging
from typing import Optional, Tuple, Any
# Configure logging for error tracking
logging.basicConfig(
    level=logging.INFO,
    format='%(asctime)s - %(levelname)s - %(message)s'
)
logger = logging.getLogger(__name__)
def validate_username(username: str) -> bool:
    """
    Validate username input for proper format and length.
    Args:
        username (str): The username to validate.
    Returns:
        bool: True if valid, False otherwise.
    """
    if not isinstance(username, str):
        return False
    if not username or len(username) == 0 or len(username) > 50:
        return False
    # Allow alphanumeric, underscore, and hyphen only
    return username.replace('_', '').replace('-', '').isalnum()
def get_user(username: str) -> Optional[Tuple[Any, ...]]:
    """
    Retrieve user information from the database using secure parameterized queries.
    Args:
        username (str): The username to search for.
    Returns:
        Optional[Tuple]: User record if found, None otherwise.
    Raises:
        ValueError: If username validation fails.
        sqlite3.Error: If database operation fails.
    """
    # Input validation
    if not validate_username(username):
        raise ValueError(
            "Invalid username: must be 1-50 characters, "
            "alphanumeric with underscore or hyphen only."
        )
    try:
        # Use context manager for automatic connection closing
        with sqlite3.connect("users.db") as conn:
            cursor = conn.cursor()
            query = "SELECT * FROM users WHERE username = ?"
            cursor.execute(query, (username,))
            result = cursor.fetchone()
            return result
    except sqlite3.Error as e:
        logger.error(f"Database error: {e}")
        raise
user_input = input("Enter username: ")
print(get_user(user_input))
```

Security Comparison

Feature	Original Code	Secure Version
SQL Injection Protection	No	Yes
Input Validation	No	Yes
Exception Handling	No	Yes
Resource Safety	Weak	Strong

Observation:
The original script was vulnerable to SQL injection due to unsafe string concatenation and lacked proper input validation and error handling. After refactoring, parameterized queries were implemented to prevent SQL injection attacks. Input validation and structured exception handling were added to improve security and reliability. The final version is secure, robust, and suitable for production deployment.

Task 5: AI-Based Automated Code Review Report
Scenario
Your team uses AI tools to perform automated preliminary code reviews before human review to improve code quality and consistency across projects.
Expected Outcome
• AI-generated review report including:
    o Identified quality issues
    o Risk areas
    o Code smell detection
    o Improvement suggestions
• Optional improved version of the code
• Demonstration of AI as a code reviewer, not just a code generator

Poorly Written Code:

```
import math

def calc(a,b,c):
 d=a+b+c
 if d>100:
  print("big")
 else:
  print("small")
 return d

x=10
y=20
z=30
r=calc(x,y,z)
print(r)
```

Refactored Code:

```python
"""
Numeric validation and classification module.
This module provides functionality to calculate the sum of numeric values
and classify them based on predefined thresholds.
"""
from typing import Union
# Configuration constants
SUM_THRESHOLD = 100
CLASSIFICATION_LARGE = "large"
CLASSIFICATION_SMALL = "small"
def calculate_and_classify(value1: Union[int, float],
                           value2: Union[int, float],
                           value3: Union[int, float],
                           threshold: int = SUM_THRESHOLD) -> tuple:
    """
    Calculate the sum of three numeric values and classify as large or small.
    Args:
        value1 (Union[int, float]): First numeric value.
        value2 (Union[int, float]): Second numeric value.
        value3 (Union[int, float]): Third numeric value.
        threshold (int): Classification threshold. Defaults to SUM_THRESHOLD.
    Returns:
        tuple: A tuple containing (sum, classification).
    Raises:
        TypeError: If any value is not a number (int or float).
        ValueError: If threshold is not a positive integer.
    Example:
        >>> calculate_and_classify(10, 20, 30)
        (60, 'small')
        >>> calculate_and_classify(50, 40, 20)
        (110, 'large')
    """
    if not isinstance(threshold, int) or threshold <= 0:
        raise ValueError("Threshold must be a positive integer.")

    for value in [value1, value2, value3]:
        if not isinstance(value, (int, float)) or isinstance(value, bool):
            raise TypeError(
                f"Expected numeric value (int/float), got {type(value).__name__}"
            )
```

```python
    total = value1 + value2 + value3
    classification = (
        CLASSIFICATION_LARGE if total > threshold
        else CLASSIFICATION_SMALL
    )
    return total, classification
def display_result(total: Union[int, float], classification: str) -> None:
    """
    Display calculation results in a user-friendly format.
    Args:
        total (Union[int, float]): The calculated sum.
        classification (str): The classification result.
    """
    print(f"Sum: {total}")
    print(f"Classification: {classification}")
def main():
    """Execute the main program logic."""
    try:
        # Input values
        first_value = 10
        second_value = 20
        third_value = 30
        # Calculate and classify
        total, classification = calculate_and_classify(
            first_value,
            second_value,
            third_value
        )
        # Display results
        display_result(total, classification)
    except TypeError as e:
        print(f"Type Error: {e}")
    except ValueError as e:
        print(f"Value Error: {e}")
    except Exception as e:
        print(f"Unexpected error: {e}")
if __name__ == "__main__":
    main()
```

Review:

Code Quality Analysis Report for task-5.py:

Issues Found and Fixed:

Issue Category	Severity	Problem	Solution
Naming Conventions	HIGH	`calc()`, `a,b,c`, `d`, `x,y,z`, `r`	Used descriptive names: calculate_and_classify(), value1, total, first_value
Code Readability	HIGH	Vague function purpose and magic numbers	Separated concerns, added meaningful output, constants for thresholds
Formatting/Style	HIGH	1-space indentation, no type hints, missing spaces	PEP 8 compliant: 4-space indents, type hints, proper spacing
Error Handling	CRITICAL	No validation, crashes on invalid input	Added type checking, ValueError handling, appropriate error messages
Documentation	HIGH	No docstrings, unused import (math)	Module docstring, comprehensive function docstrings, removed unused imports
Maintainability	HIGH	Mixed concerns (calculation + printing), magic numbers	Separated functions, configurable constants, single responsibility principle

Key Improvements:

1. **Naming**: Clear, descriptive names following snake_case convention
2. **Separation of Concerns**: Split into calculate_and_classify() and display_result() functions
3. **Type Hints**: Added Union types for numeric values
4. **Configurable Constants**: SUM_THRESHOLD and classifications are now constants
5. **Robust Error Handling**: Validates types, ranges, and provides meaningful messages
6. **PEP 8 Compliance**: Proper indentation, spacing, line length
7. **Documentation**: Module, function docstrings with Args, Returns, Raises, and Examples
8. **Testability**: Pure functions without side effects make unit testing easier
9. **Flexibility**: Function accepts optional threshold parameter
10. **Best Practices**: Uses context managers pattern, structured exception handling, main() function

Observation:

The AI-based code review successfully identified key quality issues such as poor naming conventions, lack of documentation, formatting inconsistencies, and weak error handling. It highlighted risk areas like magic numbers, mixed responsibilities, and maintainability concerns. The structured report provided clear improvement suggestions, demonstrating how AI can act as an effective preliminary reviewer before human evaluation. Overall, the process improved code clarity, consistency, and long-term maintainability.